

SECTION 07 52 00

MODIFIED BITUMINOUS MEMBRANE ROOFING

Riverside Mixed-Use Development
Project No. RMD-2025-001

PART 1 - GENERAL

1.01 REFERENCES

- ASTM D4586 - Asphalt Roof Cement
- ASTM D6162 - SBS Modified Bituminous Sheet Materials
- ASTM D6163 - APP Modified Bituminous Sheet Materials
- FM Global Loss Prevention Data Sheet 1-29
- NRCA Roofing Manual: Membrane Roof Systems
- ASTM C1289 - Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board

1.02 SYSTEM DESCRIPTION

- A. Modified bituminous membrane roofing system with minimum 20-year warranty.
- B. Total system thickness: minimum 240 mils.
- C. Wind uplift rating: FM 1-90 minimum.
- D. Fire classification: UL Class A.

1.03 WARRANTY

- A. Manufacturer's 20-year NDL (No Dollar Limit) warranty including materials and labor.
- B. Warranty to cover leaks and material defects.
- C. Installer must be manufacturer-approved applicator.

PART 2 - PRODUCTS

2.01 MEMBRANE

- A. Base sheet: ASTM D6163, Type I, Grade S. Siplast Paradiene 20 or approved equal.
- B. Cap sheet: ASTM D6162, granule-surfaced. Siplast Paradiene 30 or approved equal.

2.02 INSULATION

Layer	Material	Thickness	R-Value	Standard
Bottom	Polyiso	3 inch	R-17.4	ASTM C1289
Top	Polyiso	2.5 inch	R-14.5	ASTM C1289
Cover Board	DensDeck Prime	1/4 inch	N/A	ASTM C1177

Total assembly R-value: minimum R-30.

2.03 FLASHING AND ACCESSORIES

- A. Metal flashing: 24 ga galvanized steel, prefinished.
- B. Membrane flashing: reinforced modified bitumen, self-adhering.
- C. Sealant: polyurethane, ASTM C920, Type S, Grade NS, Class 25.
- D. Drains: cast iron, Zurn Z100 or approved equal, 4-inch diameter.

PART 3 - EXECUTION

3.01 EXAMINATION

- A. Verify deck is clean, dry, and free of projections.
- B. Confirm positive drainage slope: minimum 1/4 inch per foot.

3.02 INSULATION INSTALLATION

- A. Install in minimum two layers with staggered joints.
- B. Mechanically fasten bottom layer; adhere top layer.
- C. Taper insulation for positive drainage where required.

3.03 MEMBRANE APPLICATION

- A. Apply per manufacturer's instructions and FM Global requirements.
- B. Do not apply when ambient temperature below 40 deg F.
- C. Torch-applied method with appropriate safety precautions.
- D. Lap seams: minimum 4 inches side lap, 6 inches end lap.

3.04 QUALITY CONTROL

- A. Flood test completed areas for 48 hours minimum.
- B. Electronic leak detection per ASTM D7877.
- C. Core cuts at 5 locations per 10,000 SF.
- D. Infrared scan after 30 days to detect moisture infiltration.

END OF SECTION 07 52 00